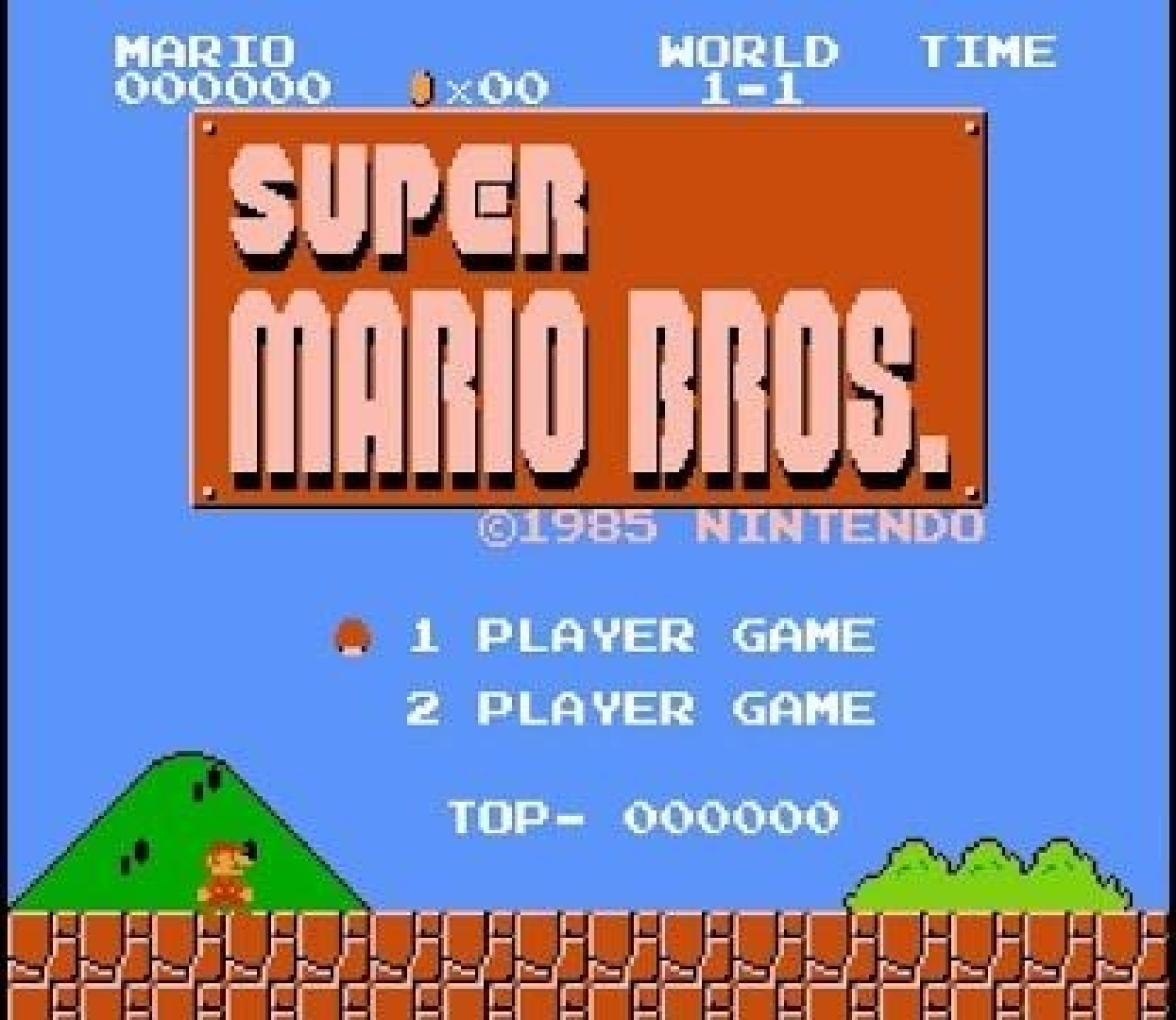
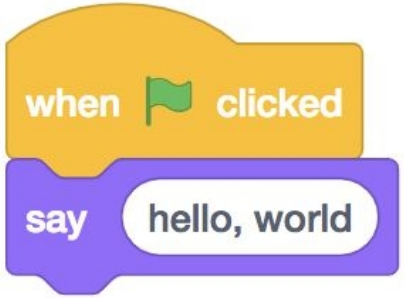
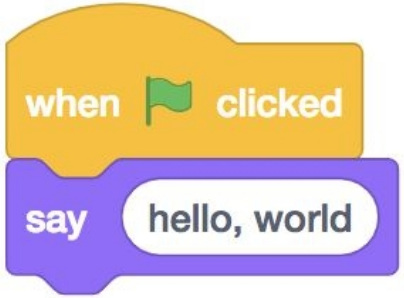
Це CS50:
Основи програмування


MARIO
000000
×00
WORLD
1-1
TIME
SUPER
MARIO BROS.
©1985 NINTENDO
1 PLAYER GAME
2 PLAYER GAME
TOP- 000000


when clicked
say
hello, world




```
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

```c
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

```c
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

```
01111111 01000101 01001100 01000110 00000010 00000001 00000001 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000010 00000000 00111110 00000000 00000001 00000000 00000000 00000000
10110000 00000101 01000000 00000000 00000000 00000000 00000000 00000000
01000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
11010000 00010011 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 01000000 00000000 00111000 00000000
00001001 00000000 01000000 00000000 00100100 00000000 00100001 00000000
00000110 00000000 00000000 00000000 00000101 00000000 00000000 00000000
01000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
01000000 00000000 01000000 00000000 00000000 00000000 00000000 00000000
01000000 00000000 01000000 00000000 00000000 00000000 00000000 00000000
11111000 00000001 00000000 00000000 00000000 00000000 00000000 00000000
11111000 00000001 00000000 00000000 00000000 00000000 00000000 00000000
00001000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000011 00000000 00000000 00000000 00000100 00000000 00000000 00000000
00111000 00000010 00000000 00000000 00000000 00000000 00000000 00000000
...
```

```
clang hello.c

./a.out
```

```
clang -o hello hello.c

./hello
```

```
make hello

./hello
```

```c
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

```
#include <cs50.h>
#include <stdio.h>

int main(void)
{
    string name = get_string("Name: ");
    printf("hello, %s\n", name);
}
```

```
clang hello.c

./a.out
```

```
clang hello.c -lcs50

./a.out
```

```
clang -o hello hello.c -lcs50

./hello
```

```
make hello

./hello
```

компіляція

препроцессінг

компіляція

збирання

компонування

препроцессінг

компіляція

збирання

компонування

```
#include <cs50.h>
#include <stdio.h>


int main(void)
{
    string name = get_string("Name: ");
    printf("hello, %s\n", name);
}
```

```c
#include <cs50.h>
#include <stdio.h>


int main(void)
{
    string name = get_string("Name: ");
    printf("hello, %s\n", name);
}
```

```
string get_string(string prompt);
#include <stdio.h>


int main(void)
{
    string name = get_string("Name: ");
    printf("hello, %s\n", name);
}
```

```
string get_string(string prompt);
#include <stdio.h>


int main(void)
{
    string name = get_string("Name: ");
    printf("hello, %s\n", name);
}
```

```
string get_string(string prompt);
int printf(const char *format, ...);


int main(void)
{
    string name = get_string("Name: ");
    printf("hello, %s\n", name);
}
```

```
...
string get_string(string prompt);
int printf(const char *format, ...);
...

int main(void)
{
    string name = get_string("Name: ");
    printf("hello, %s\n", name);
}
```

препроцессінг

компіляція

збирання

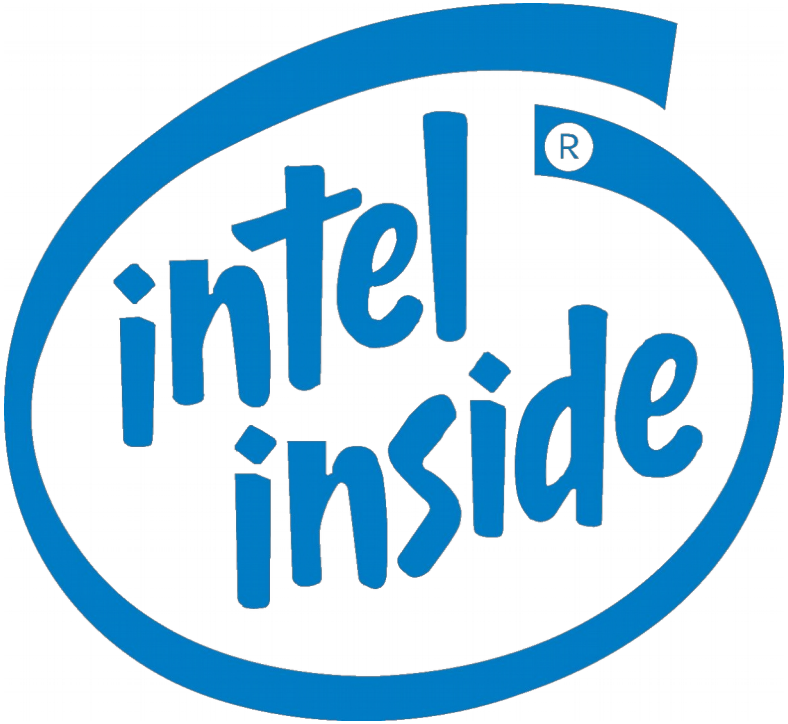
компонування

```
...
string get_string(string prompt);
int printf(const char *format, ...);
...

int main(void)
{
    string name = get_string("Name: ");
    printf("hello, %s\n", name);
}
```

```
...
main:                                   # @main
    .cfi_startproc
# BB#0:
    pushq    %rbp
.Ltmp0:
    .cfi_def_cfa_offset 16
.Ltmp1:
    .cfi_offset %rbp, -16
    movq    %rsp, %rbp
.Ltmp2:
    .cfi_def_cfa_register %rbp
    subq    $16, %rsp
    xorl    %eax, %eax
    movl    %eax, %edi
    movabsq    $.L.str, %rsi
    movb    $0, %al
    callq    get_string
    movabsq    $.L.str.1, %rdi
    movq    %rax, -8(%rbp)
    movq    -8(%rbp), %rsi
    movb    $0, %al
    callq    printf
    ...
```

```
...
main:                                   # @main
    .cfi_startproc
# BB#0:
    pushq    %rbp
.Ltmp0:
    .cfi_def_cfa_offset 16
.Ltmp1:
    .cfi_offset %rbp, -16
    movq    %rsp, %rbp
.Ltmp2:
    .cfi_def_cfa_register %rbp
    subq    $16, %rsp
    xorl    %eax, %eax
    movl    %eax, %edi
    movabsq    $.L.str, %rsi
    movb    $0, %al
    callq    get_string
    movabsq    $.L.str.1, %rdi
    movq    %rax, -8(%rbp)
    movq    -8(%rbp), %rsi
    movb    $0, %al
    callq    printf
    ...
```

```
...
main:                                   # @main
    .cfi_startproc
# BB#0:
    pushq    %rbp
.Ltmp0:
    .cfi_def_cfa_offset 16
.Ltmp1:
    .cfi_offset %rbp, -16
    movq    %rsp, %rbp
.Ltmp2:
    .cfi_def_cfa_register %rbp
    subq    $16, %rsp
    xorl    %eax, %eax
    movl    %eax, %edi
    movabsq    $.L.str, %rsi
    movb    $0, %al
    callq    get_string
    movabsq    $.L.str.1, %rdi
    movq    %rax, -8(%rbp)
    movq    -8(%rbp), %rsi
    movb    $0, %al
    callq    printf
    ...
```


intel
inside
R

препроцессінг

компіляція

збирання

компонування

```
...
main:                                   # @main
    .cfi_startproc
# BB#0:
    pushq    %rbp
.Ltmp0:
    .cfi_def_cfa_offset 16
.Ltmp1:
    .cfi_offset %rbp, -16
    movq    %rsp, %rbp
.Ltmp2:
    .cfi_def_cfa_register %rbp
    subq    $16, %rsp
    xorl    %eax, %eax
    movl    %eax, %edi
    movabsq    $.L.str, %rsi
    movb    $0, %al
    callq    get_string
    movabsq    $.L.str.1, %rdi
    movq    %rax, -8(%rbp)
    movq    -8(%rbp), %rsi
    movb    $0, %al
    callq    printf
    ...
```

```
01111111010001010100110001000110
00000010000000010000000100000000
00000000000000000000000000000000
00000000000000000000000000000000
00000001000000000011111000000000
00000001000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
10100000000000100000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
01000000000000000000000000000000
00000000000000001000000000000000
00001010000000000000000100000000
01010101010010001000100111100101
01001000100000111110110000010000
00110001110000001000100111000111
01001000101111100000000000000000
00000000000000000000000000000000
00000000000000001011000000000000
11101000000000000000000000000000
00000000010010001011111100000000
00000000000000000000000000000000
00000000000000000000000001001000
...
```

препроцессінг

компіляція

збирання

компонування

hello.c

hello.c

cs50.c

hello.c

cs50.c

stdio.c

hello.c

cs50.c

printf.c

```
01111111010001010100110001000110
00000010000000010000000100000000
00000000000000000000000000000000
00000000000000000000000000000000
00000001000000000011111000000000
00000001000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
10100000000000100000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
01000000000000000000000000000000
00000000000000001000000000000000
00001010000000000000000100000000
01010101010010001000100111100101
01001000100000111110110000010000
00110001110000001000100111000111
01001000101111100000000000000000
00000000000000000000000000000000
00000000000000001011000000000000
11101000000000000000000000000000
00000000010010001011111100000000
00000000000000000000000000000000
00000000000000000000001001000
...
```

cs50.c

printf.c

```
01111111010001010100110001000110   01111111010001010100110001000110
00000010000000010000000100000000   00000010000000010000000100000000
00000000000000000000000000000000   00000000000000000000000000000000
00000000000000000000000000000000   00000000000000000000000000000000
00000001000000000011111000000000   00000011000000000011111000000000
00000001000000000000000000000000   00000001000000000000000000000000
00000000000000000000000000000000   11000000000011110000000000000000
00000000000000000000000000000000   00000000000000000000000000000000
00000000000000000000000000000000   01000000000000000000000000000000
00000000000000000000000000000000   00000000000000000000000000000000
10100000000001000000000000000000   00101000001100100000000000000000
00000000000000000000000000000000   00000000000000000000000000000000
00000000000000000000000000000000   00000000000000000000000000000000
01000000000000000000000000000000   01000000000000000011100000000000
00000000000000001000000000000000   00000111000000001000000000000000
00001010000000000000000100000000   00011100000000000001100100000000
01010101010010001000100111100101   00000001000000000000000000000000
01001000100000111110110000010000   00000101000000000000000000000000
00110001110000001000100111000111   00000000000000000000000000000000
01001000101111100000000000000000   00000000000000000000000000000000
00000000000000000000000000000000   00000000000000000000000000000000
00000000000000001011000000000000   00000000000000000000000000000000
11101000000000000000000000000000   00000000000000000000000000000000
00000000010010001011111100000000   00000000000000000000000000000000
00000000000000000000000000000000   01011100001001010000000000000000
00000000000000000000000001001000   00000000000000000000000000000000
...                                ...
```

printf.c

```
011111110100010101001100010001100
000000100000000010000000100000000
000000000000000000000000000000000
000000000000000000000000000000000
000000010000000000111110000000000
000000010000000000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
101000000000001000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
010000000000000000000000000000000
000000000000000010000000000000000
000010100000000000000000100000000
010101010100100010001001111001 01
010010001000001111101100000100 00
001100011100000010001001110001 11
010010001011111000000000000000000
000000000000000000000000000000000
000000000000000010110000000000000
111010000000000000000000000000000
000000000100100010111111000000000
000000000000000000000000000000000
000000000000000000000000001001000
...
```

```
011111110100010101001100010001100
000000100000000010000000100000000
000000000000000000000000000000000
000000000000000000000000000000000
000000110000000000111110000000000
000000010000000000000000000000000
110000000000011110000000000000000
000000000000000000000000000000000
010000000000000000000000000000000
000000000000000000000000000000000
001010000011001000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
010000000000000000011100000000000
000001110000000001000000000000000
000111000000000000011001000000000
000000010000000000000000000000000
000001010000000000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
010111000010010100000000000000000
000000000000000000000000000000000
```

```
001011110110110001101001011000 10
011000110010111001110011011011 11
001011100011011000100000001011 11
011101010111001101110010001011 11
011011000110100101100010001011 11
011110000011100000110110010111 11
001101100011010000101101011011 00
011010010110111001110101011110 00
001011010110011101101110011101 01
001011110110110001101001011000 10
011000110101111101101110011011 11
011011100111001101101000011000 01
011100100110010101100100001011 10
011000010010000000100000010000 01
010100110101111101001110010001 01
010001010100010001000101010001 00
001000000010100000100000000101 11
011011000110100101100010001011 11
011110000011100000110110010111 11
001101100011010000101101011011 00
011010010110111001110101011110 00
001011010110011101101110011101 01
001011110110110001100100001011 01
011011000110100101101110011101 01
011110000010110101111000001110 00
001101100010110100110110001101 00
```

```
...
```

0111111101000101010011000100011000000100000000100000001000000000000000000000000000000000000000000000
0000000000000000000000000000000100000000001111100000000000000100000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000101000000000001000000000000000000000000000000000000000000000000000000000000000000000000
0000000001000000000000000000000000000000000000000000000000000100000000000000000101000000000000000100000
0001010101010010001000100111100101010010001000001111101100000100000011000111000000100010011100011101000
1000101111100000000000000000000000000000000000000000000000000000000000000000000000000101100000000000111010000
0000000000000000000000000000000100100010111111000000000000000000000000000000000000000000000000000000000
0000000001001000...0111111101000101010011000100011000000100000000100000001000000000000000000000000000
0000000000000000000000000000000000000000000000000011000000000011111000000000000000000100000000000000000
0000000110000000000111100000000000000000000000000000000000000000000000000000000100000000000000000000000
0000000000000000000000000000000000000101000001100100000000000000000000000000000000000000000000000000000
0000000000000000000000000000000100000000000000011100000000000000011100000000010000000000000000011100000
0000000011001000000000000000100000000000000000000000000000101000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000

Intel® Core™ i7 processor

help50

printf

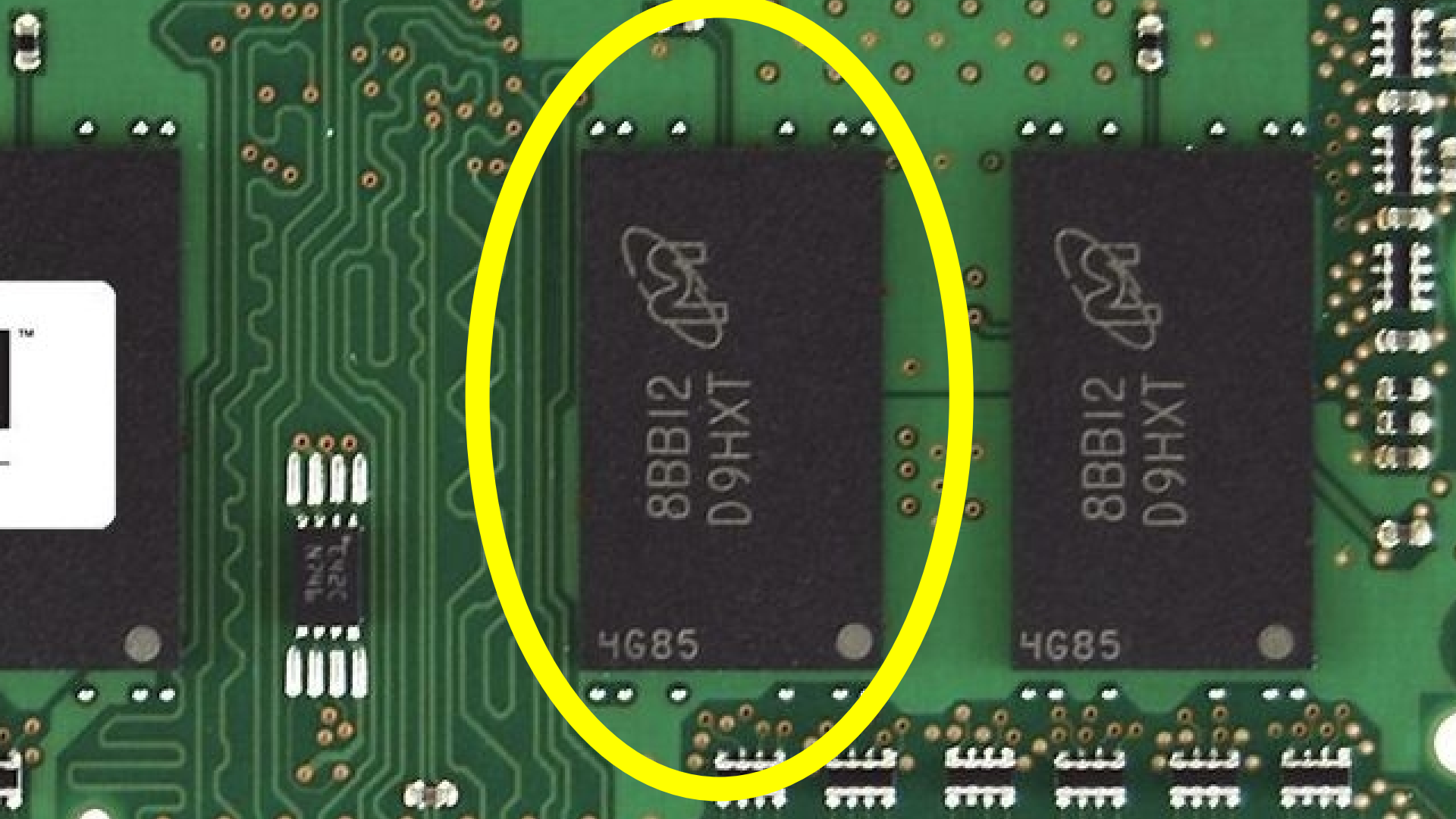
style50

crucial
by Micron
47J6
4G85
8BB12
D9HXT
4G85
8BB12
D9HXT
4G85

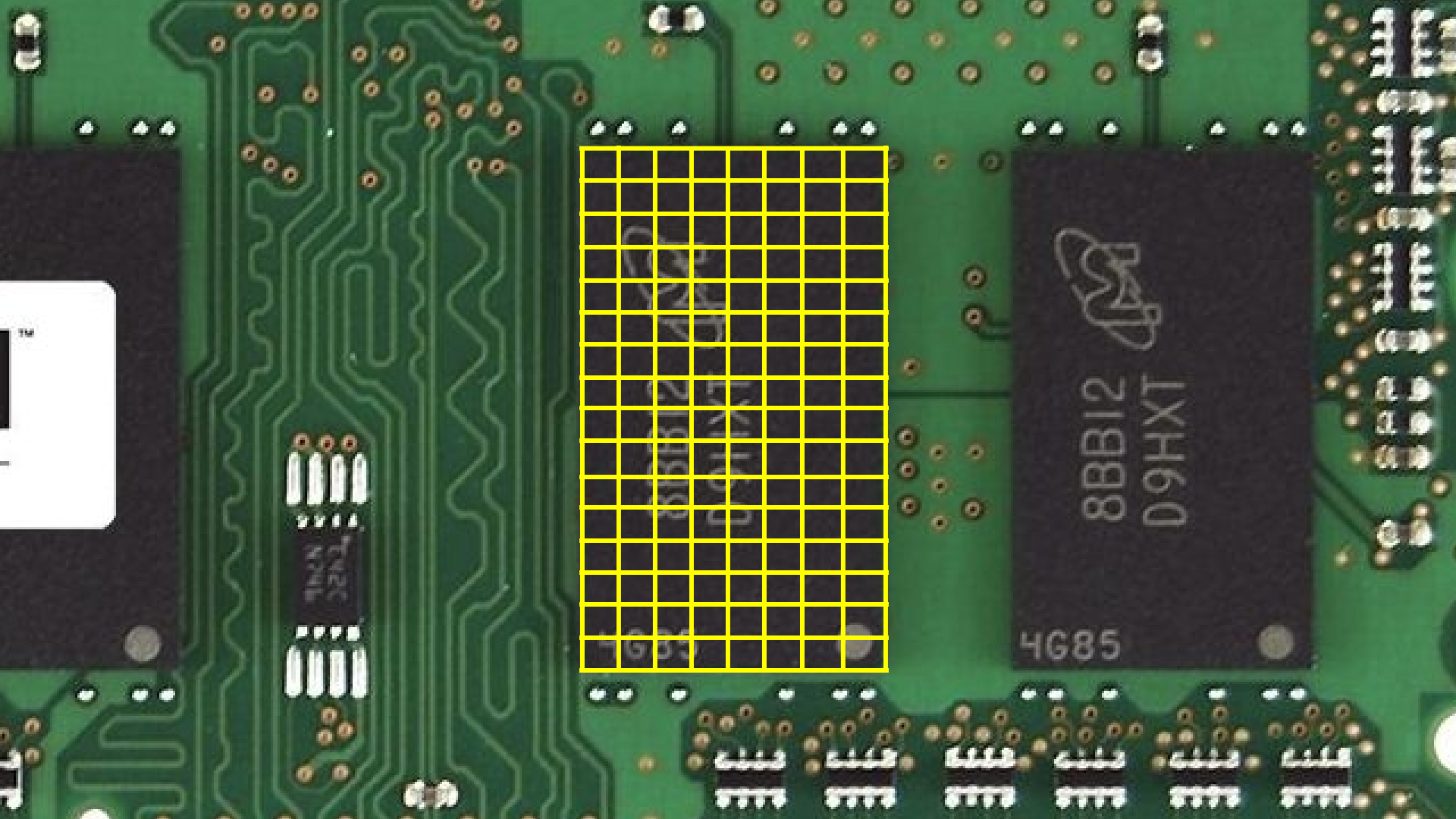
amazon.com

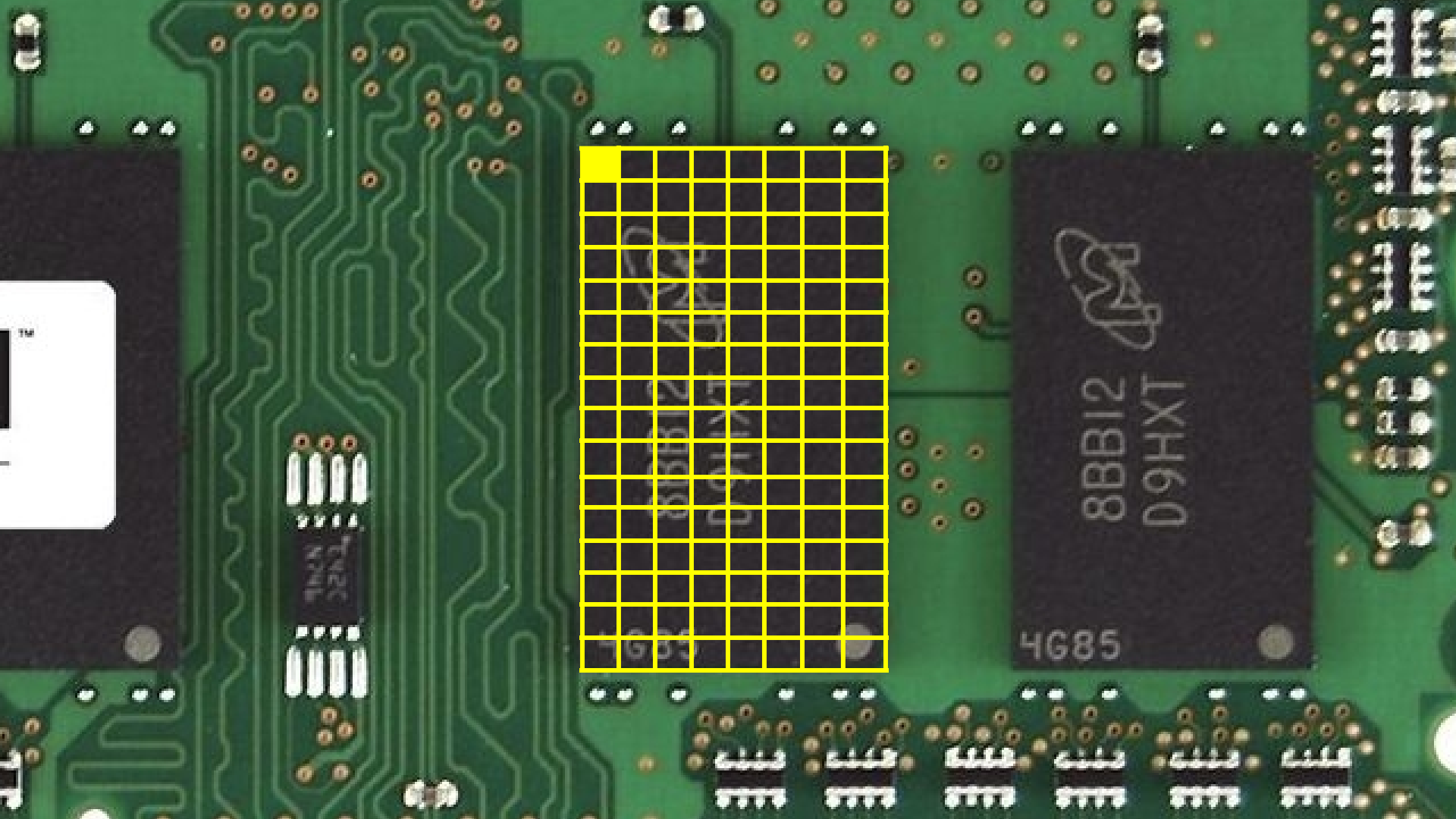
crucial
by Micron
8BB12
D9HXT
8BB12
D9HXT
47J6
4G85
4G85
4G85

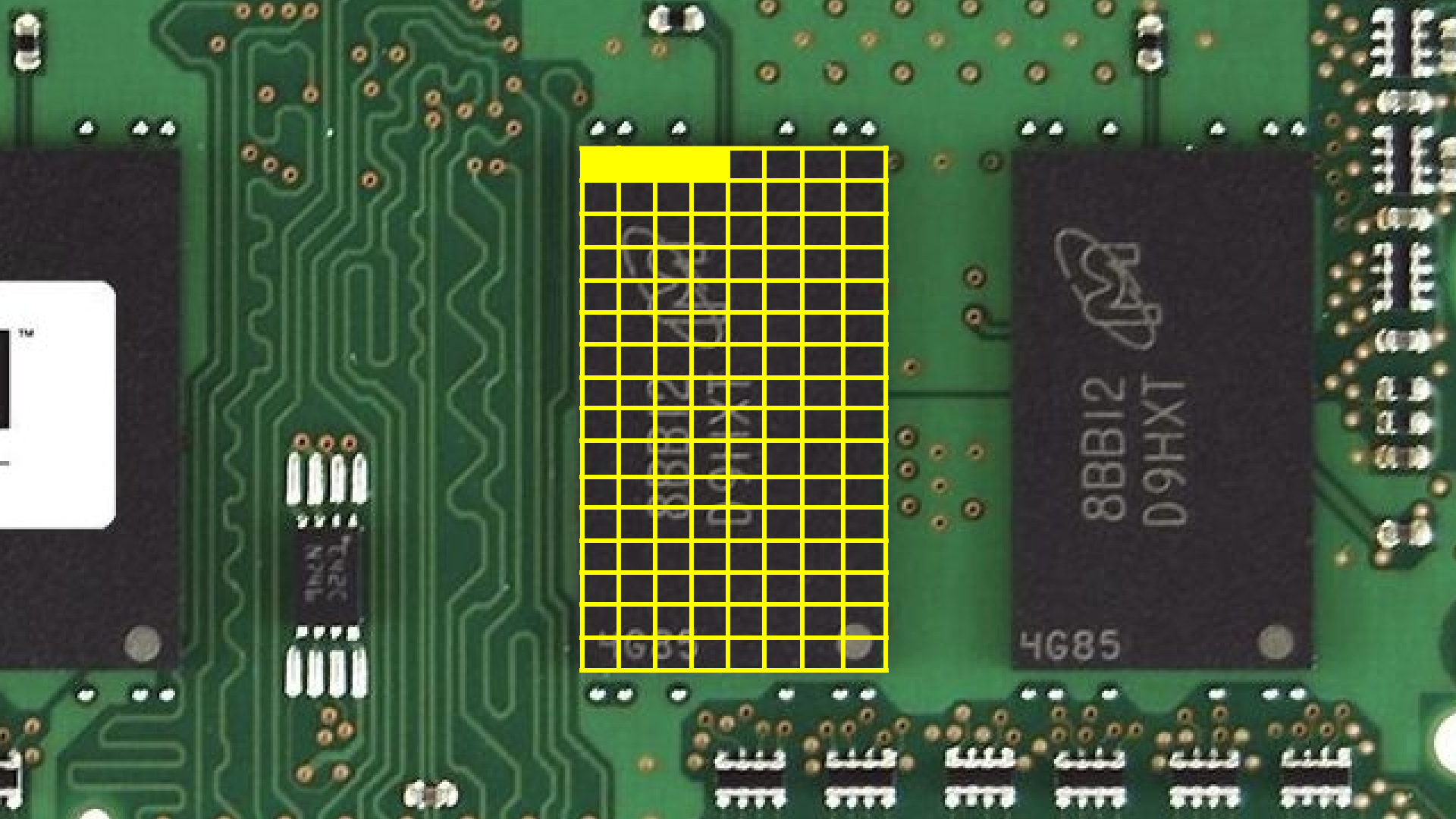
amazon.com

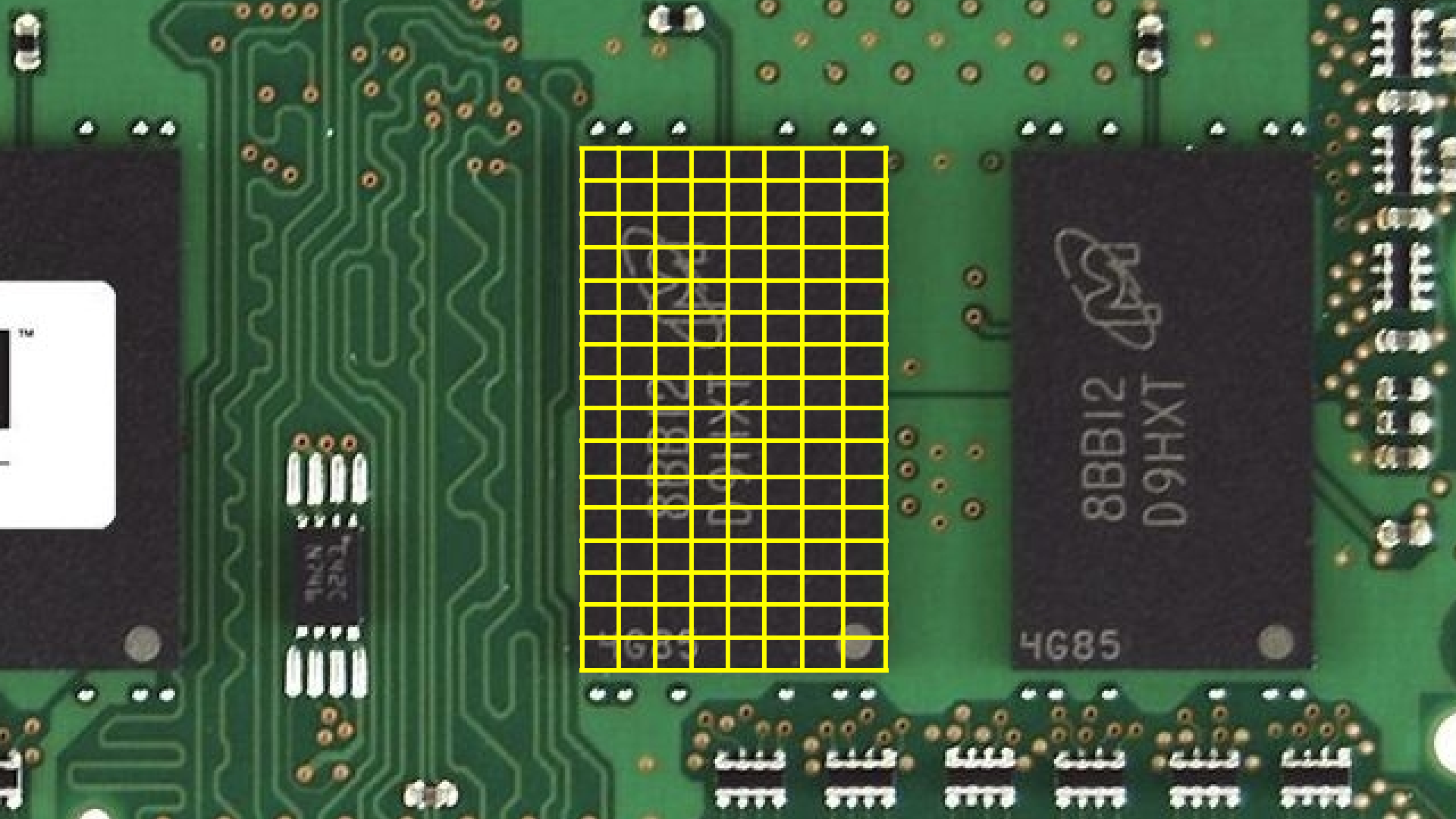

8BBI2
D9HXT
4G85
8BBI2
D9HXT
4G85

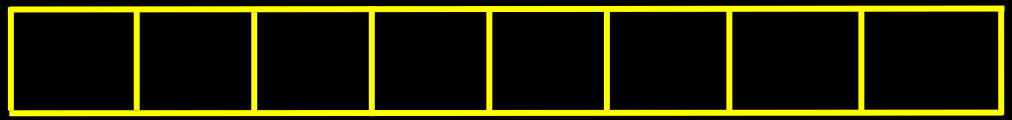
8BB12
D9HXT
4G85
8BB12
D9HXT
4G85


8BB12
D9HXT
4G85


8BB12
D9HXT
4G85


8BB12
D9HXT
4G85


8BB12
D9HXT
4G85

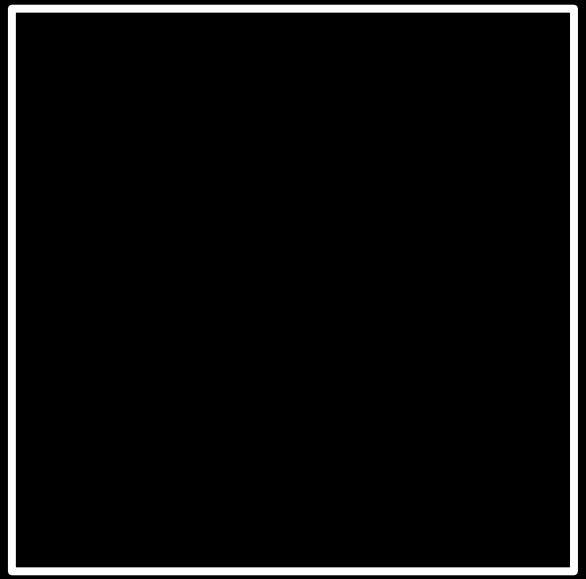


масив

рядок

вхідні дані→ → вихідні дані

звичайний
текст →



→ зашифро-
ваний текст

H I !

72 73 33

CLASS OF SERVICE DESIRED	
Fast Day Message	X
Day Letter	
Night Message	
Night Letter	

Patrons should mark an X opposite the class of service desired; OTHERWISE THE TELEGRAM WILL BE TRANSMITTED AS A FAST DAY MESSAGE.

WESTERN UNION TELEGRAM

NEWCOMB CARLTON, PRESIDENT

CHARGE 8369

Receiver's No. MC

Check 3300

Time Filed

Send the following telegram, subject to the terms on back hereof, which are hereby agreed to

via Galveston

JAN 19 1917

862.20212/128

GERMAN LEGATION

MEXICO CITY

130 13042 13401 8501 115 3528 416 17214 6491 11310
18147 18222 21560 10247 11518 23677 13605 3494 14936
98092 5905 11311 10392 10371 0302 21290 5161 39695
23571 17504 11269 18276 18101 0317 0228 17694 4473
23284 22200 19452 21589 67893 5569 13918 8958 12137
1333 4725 4458 5905 17166 13851 4458 17149 14471 6706
13850 12224 6929 14991 7382 15857 67893 14218 36477
5870 17553 67893 5870 5454 16102 15217 22801 17138
21001 17388 7446 23638 18222 6719 14331 15021 23845
3156 23552 22096 21604 4797 9497 22464 20855 4377
23610 18140 22260 5905 13347 20420 39689 13732 20667
6929 5275 18507 52262 1340 22049 13339 11265 22295
10439 14814 4178 6992 8784 7632 7357 6926 52262 11267
21100 21272 9346 9559 22464 15874 18502 18500 15857
2188 5376 7381 98092 16127 13486 9350 9220 76036 14219
5144 2831 17920 11347 17142 11264 7667 7762 15099 9110
10482 97556 3569 3670

BERNSTORFF.

INSPECTED

Charge German Embassy.

4458	gemeinsam
17149	Friedensschluß.
14471	⊙
6706	reichlich
13850	finanziell
12224	unterstützung
6929	und
14991	einverständnis
7382	unserseits.
158(5)7	da/3
67893	Mexico:
14218	in
36477	Texas
5870	⊙
17553	neu
67893	Mexico.
5870	⊙
5454	AR
16102	IZ
15217	ON
22801	A

like a pall on sculpture, till another man took the burden from him and went up to the house with his dead.

When Mr. Raleigh entered the house again, it was at break of dawn. Some one opened the library-door and beckoned him in. Marguerite sprang into his arms.

"What if she had died?" said Mrs. Purcell, with her swift satiric breath, and folding a web of muslin over her arm. "See! I had got out the shroud. As it is, we drink *skôl* and say grace at breakfast. The funeral baked-meats shall coldly furnish forth the marriage-feast. You men are all alike. *Le Roi est mort! Vive la Reine!*"

PAUL REVERE'S RIDE.

LISTEN, my children, and you shall hear
Of the midnight ride of Paul Revere,
On the eighteenth of April, in Seventy-Five:
Hardly a man is now alive
Who remembers that famous day and year.

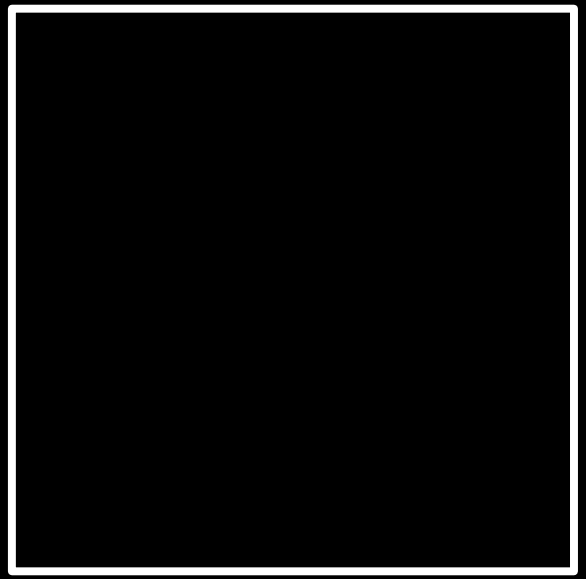
He said to his friend, — "If the British march
By land or sea from the town to-night,
Hang a lantern aloft in the belfry-arch
Of the North-Church-tower, as a signal-light, —
One if by land, and two if by sea;
And I on the opposite shore will be,
Ready to ride and spread the alarm
Through every Middlesex village and farm,
For the country-folk to be up and to arm."

Then he said good-night, and with muffled oar
Silently rowed to the Charlestown shore,
Just as the moon rose over the bay,
Where swinging wide at her moorings lay
The Somersett, British man-of-war:
A phantom ship, with each mast and spar
Across the moon, like a prison-bar,
And a huge, black hulk, that was magnified
By its own reflection in the tide.

Meanwhile, his friend, through alley and street
Wanders and watches with eager ears,
Till in the silence around him he hears
The muster of men at the barrack-door,
The sound of arms, and the tramp of feet,
And the measured tread of the grenadiers
Marching down to their boats on the shore.

Then he climbed to the tower of the church,
Up the wooden stairs, with stealthy tread,

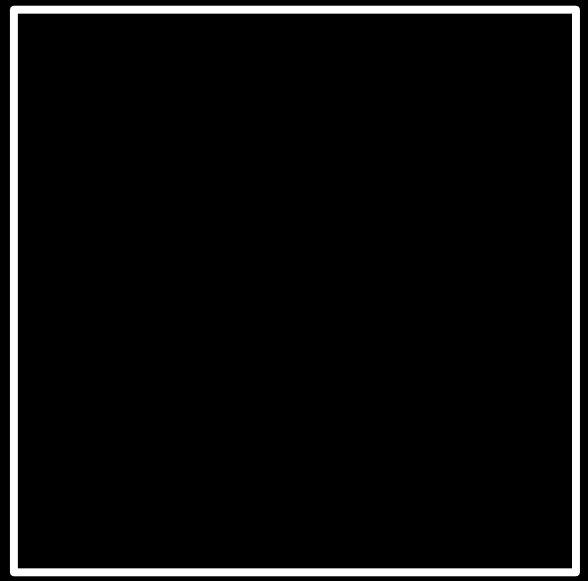
звичайний
текст →



→ зашифро-
ваний текст

ключ →

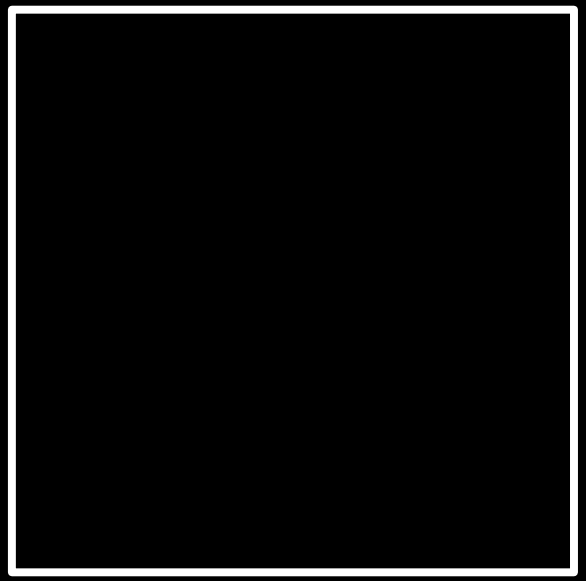
звичайний
текст →



→ зашифро-
ваний текст

1 →

звичайний
текст →



→ зашифро-
ваний текст

I L O V E Y O U

73 L O V E Y O U

73 76 0 V E Y 0 U

73 76 79 V E Y O U

73 76 79 86 E Y O U

73 76 79 86 69 Y O U

73 76 79 86 69 89 O U

73 76 79 86 69 89 79 U

73 76 79 86 69 89 79 85

74 76 79 86 69 89 79 85

74 77 79 86 69 89 79 85

74 77 80 86 69 89 79 85

74 77 80 87 69 89 79 85

74 77 80 87 70 89 79 85

74 77 80 87 70 90 79 85

74 77 80 87 70 90 80 85

74 77 80 87 70 90 80 86

J 77 80 87 70 90 80 86

J M 80 87 70 90 80 86

J M P 87 70 90 80 86

J M P W 70 90 80 86

J M P W F 90 80 86

J M P W F Z 80 86

J M P W F Z P 86

J M P W F Z P V

```
#include <stdio.h>

int main(void)
{
    ...
}
```

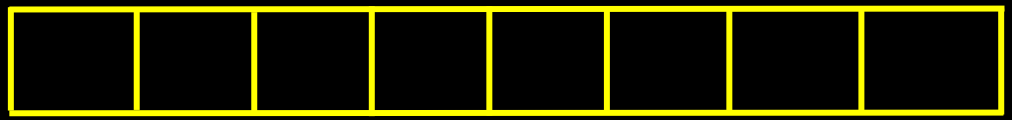
```c
#include <stdio.h>

int main(void)
{
    ...
}
```

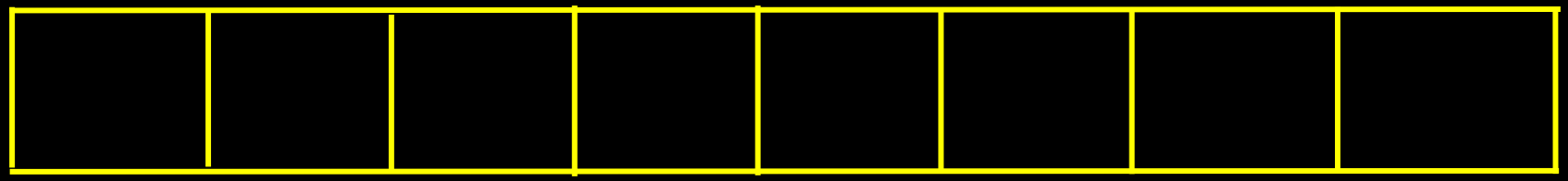
```
#include <stdio.h>

int main(int argc, string argv[])
{
    ...
}
```

```
#include <stdio.h>

int main(int argc, string argv[])
{
    ...
}
```

```
#include <stdio.h>

int main(int argc, string argv[])
{
    ...
}
```

```c
#include <stdio.h>

int main(void)
{
    ...
}
```





Сортування бульбашкою

```
repeat until no swaps
    for i from 0 to n-2
        if i'th and i+1'th elements out of order
            swap them
```

Сортування вибором

```
for i from 0 to n-1
    find smallest element between i'th and n-1'th
    swap smallest with i'th element
```

$(n - 1)$

$(n - 1) + (n - 2)$

$$(n - 1) + (n - 2) + \ldots + 1$$

$(n - 1) + (n - 2) + \ldots + 1$

$n(n - 1)/2$

$(n - 1) + (n - 2) + \ldots + 1$

$n(n - 1)/2$

$(n^2 - \mathrm{n})/2$

$(n - 1) + (n - 2) + \ldots + 1$

$n(n - 1)/2$

$(n^2 - \mathrm{n})/2$

$n^2/2 - n/2$

$(n - 1) + (n - 2) + \ldots + 1$

$n(n - 1)/2$

$(n^2 - \mathrm{n})/2$

$n^2/2 - n/2$

$O(n^2)$

$n^2/2 - n/2$

$n^2/2 - n/2$

$1{,}000{,}000^2/2 - 1{,}000{,}000/2$

$n^2/2 - n/2$

$1{,}000{,}000^2/2 - 1{,}000{,}000/2$

$500{,}000{,}000{,}000 - 500{,}000$

$n^2/2 - n/2$

$1{,}000{,}000^2/2 - 1{,}000{,}000/2$

$500{,}000{,}000{,}000 - 500{,}000$

$499{,}999{,}500{,}000$

$n^2/2 - n/2$

$1{,}000{,}000^2/2 - 1{,}000{,}000/2$

$500{,}000{,}000{,}000 - 500{,}000$

$499{,}999{,}500{,}000$

$O(n^2)$

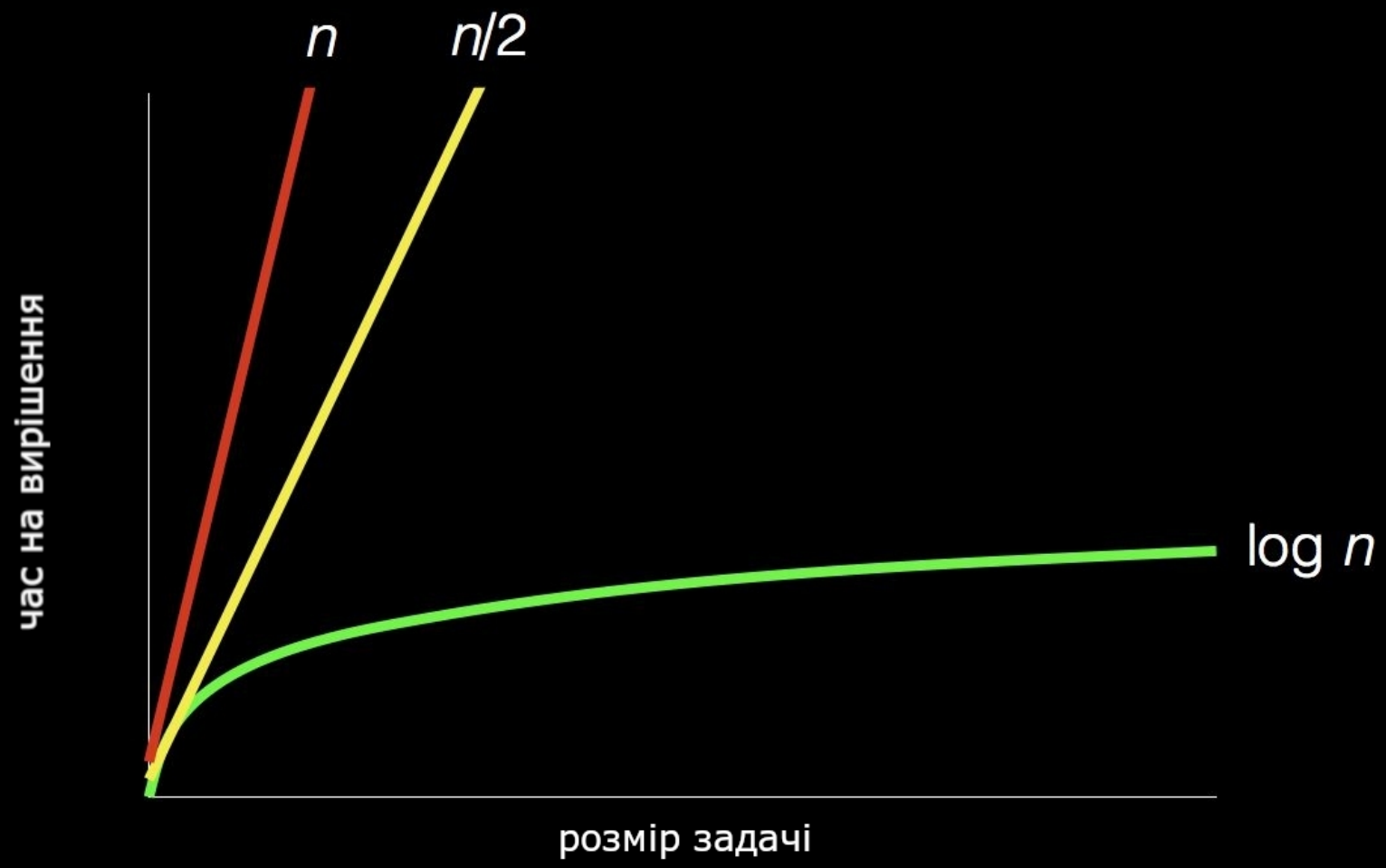
$O(n^2)$

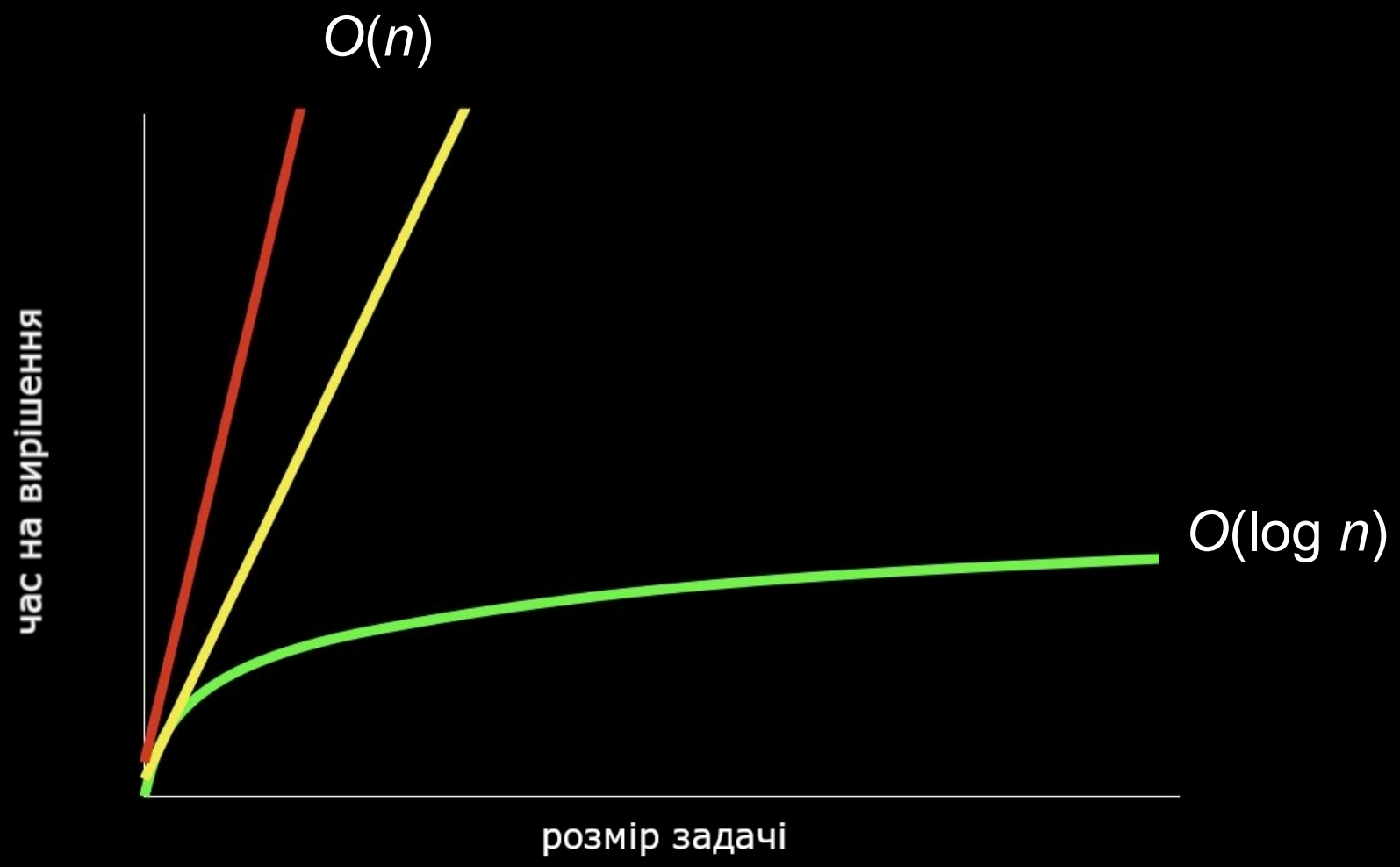
$O(n \log n)$

$O(n)$

$O(\log n)$

$O(1)$


n
n/2
log n
час на вирішення
розмір задачі


O(n)
O(log n)
час на вирішення
розмір задачі

$\Omega(n^2)$

$\Omega(n \log n)$

$\Omega(n)$

$\Omega(\log n)$

$\Omega(1)$

$\Theta(n^2)$

$\Theta(n \log n)$

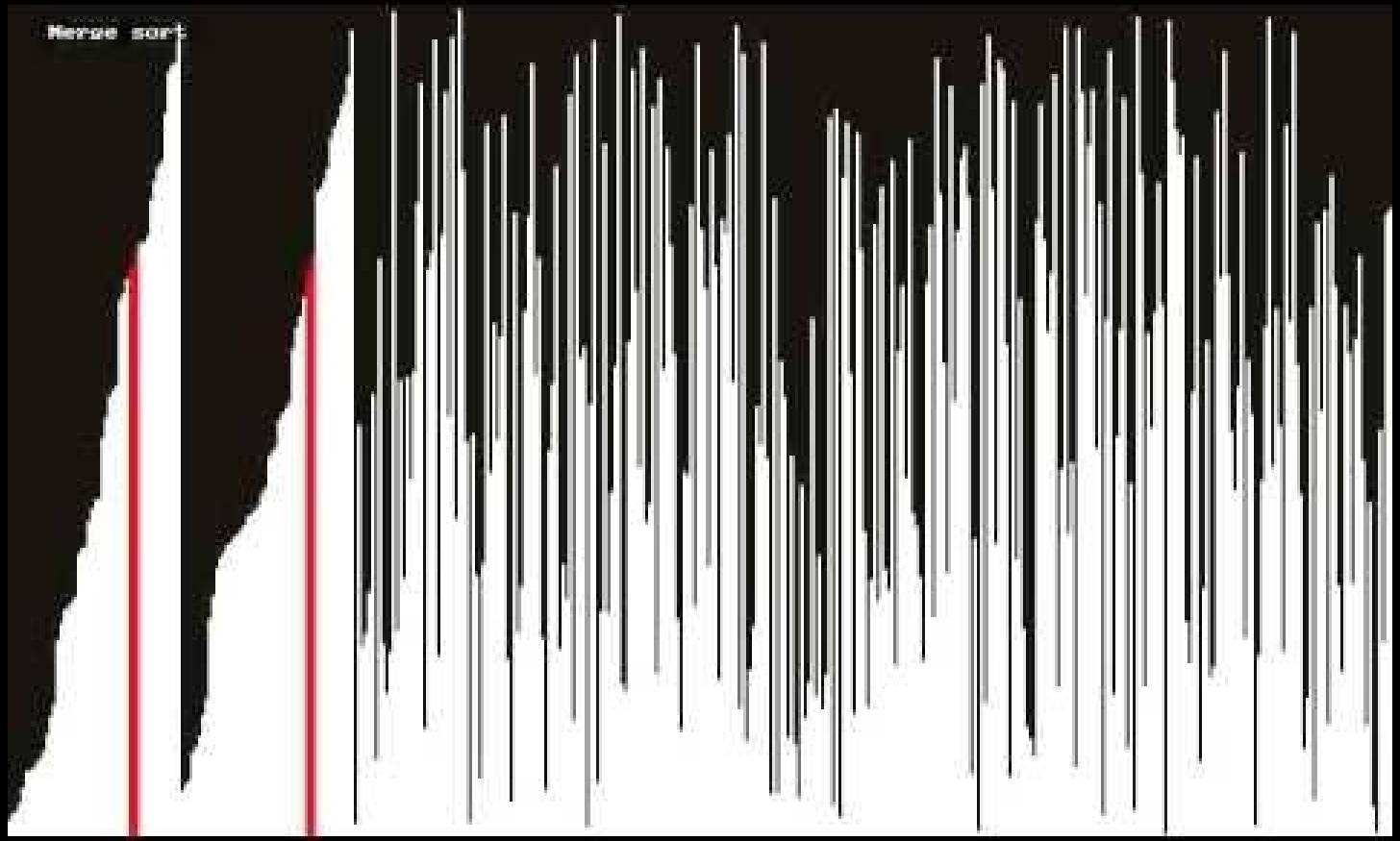
$\Theta(n)$

$\Theta(\log n)$

$\Theta(1)$

Сортування злиттям

```
on input of n elements
    if n < 2
        return
    else
        sort left half of elements
        sort right half of elements
        merge sorted halves
```


Merge sort

Це CS50:
Основи програмування